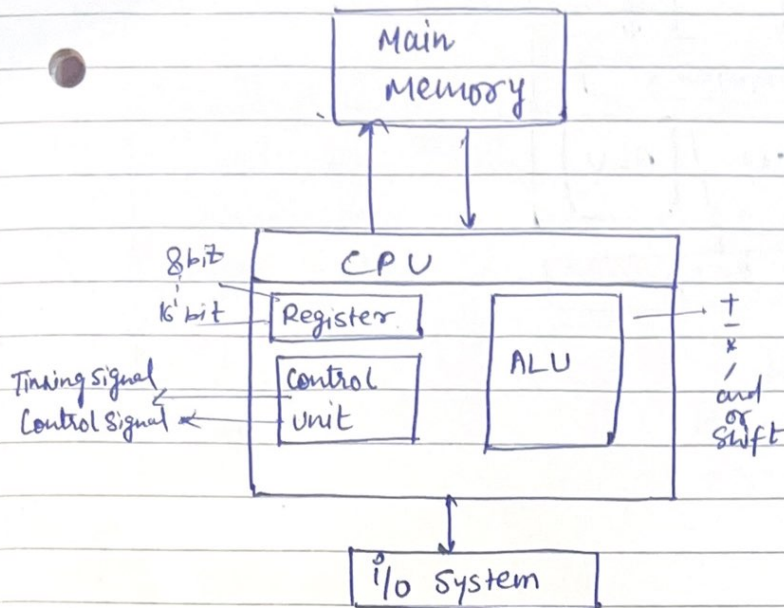


Von-Neumann Architecture



- Basically 2 Types of computers:
 - 1) fixed program computers: Specific function & Couldn't ^{reprogrammed}
 - 2) Stored program computers: Can be reprogrammed, applications are stored on it.

→ Von neumann architecture is based on stored program concept. In this stored-program concept, programs and data are stored in a ^{common} ~~separate~~ storage unit called memory.

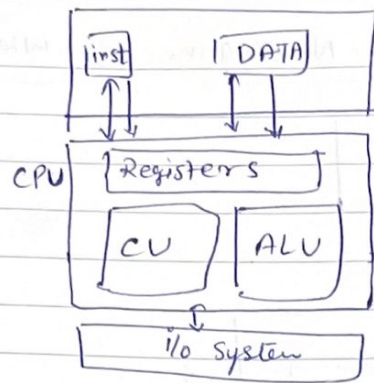
→ It is a digital computer architecture whose design is based on the concept of stored program computer where program data and instruction data are stored in the same memory.

→ designed by John von-neumann in 1945.

HARVARD ARCHITECTURE

→ is a digital computer architecture whose design is based on the concept where there are separate storage and separate buses for instructions and data.

It was basically developed to overcome the bottleneck of von-neumann architecture.



→ Diff. B/w Harvard and Von-neumann architecture.

BASIC UNITS OF COMPUTERS:

- | | | |
|---|---|---|
| 1) Main Memory | 2) CPU | 3) I/O Devices |
| generally used to store the data. | Used for instruction Execution of Computer Program. | I/O device takes the input from user and give O/P after processing. |
| divided into 2-parts
• Primary • Secondary | 3 major components
• CU • ALU • Registers | |

CPU: 1) CU → handles all processor control signals, it directs all input O/P flows, fetches code for instⁿ, and controls how data move around the system.

2) ALU → Responsible for all Arithmetic and Logical operations.

3) Registers → it refers to high speed storage areas in CPU. The data processed by the CPU are fetched from registers.

Types → 1) Accumulator 2) Program Counter 3) Memory Address Reg.
4) memory data register 5) Current Instruction Register
6) Instruction Buffer register.

* Buses → Data transmission from one part to another part of the computer.

1) Data Bus 2) Address Bus 3) Control Bus.

Fixed and Floating numbers:

- Fixed point representation has fixed number of bits for integers part and for fractional part.
- floating point refers to the fact that a number's radix point can float, that is it can be placed anywhere relative to the significant digits of the numbers.

fixed point

101.00

23.53

floating point

101.00 = 10.100×2^0

23.53 = 235.3×10^{-1}

* fixed point Representation:

$$\begin{aligned} & 11010.1 \rightarrow \text{fixed } 26.17 \\ \Rightarrow & 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} \\ \Rightarrow & 16 + 8 + 0 + 2 + 0 + .5 \\ \Rightarrow & \underline{26.5} \end{aligned}$$

- A number is always divided by two when the bit pattern of the number is shifted to the right by one bit.
- A number is multiplied by two when it is moved left one bit.

→ There are two types of floating point representation

- 1) Unsigned
- 2) Signed.

↓ example given

* Signed representation: Signed Binary numbers (+ve and -ve)

can be represented in one of three ways:

- a) Signed - Magnitude form
- b) 1's Complement form
- c) 2's Complement form

a) Signed-Magnitude form: Here the number's sign is represented by the MSB (also called leftmost bit) while its magnitude is shown by remaining bits. [Considering 8-bit representation]

Eg

$$\begin{aligned} 120 &= 00111000 \\ -120 &= 10111000 \end{aligned}$$

b) 1's Complement form: when a positive number is complemented by 1, then the result is a negative number.
 → Similarly Complementing a ~~positive~~^{-ve} number by 1, the Result is a positive number

$$120 = 00111000$$

$$-120 = 11000111 \rightarrow 1's \text{ complement}$$

c) 2's Complement form: By adding one to the signed binary number's 1's complement, a binary number can be converted to its 2's complement. So the 2's complement of a positive number, results in a negative number.
 The complement of a negative number by 2 derive a +ve number.

$$120 = 00111000$$

$$11000111 \text{ 1's complement}$$

$$+1$$

$$-120 = 11001000 \rightarrow 2's \text{ complement}$$

Eg $2.5 = 010.1$

$$010.1 \rightarrow +2.5$$

$$101.0 \rightarrow 1's \text{ complement}$$

$$+1$$

$$101.1 \rightarrow -2.5 [2's \text{ complement}]$$

* The 1's Complement Range is -127 to 127

* The 2's Complement Range is -128 to 127

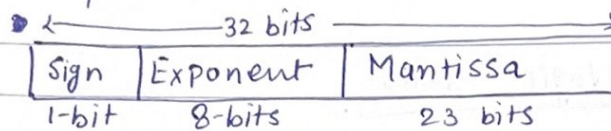
Floating point Representation

It has three part: Mantissa, Base, Exponent

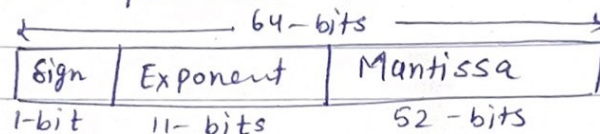
Number	Mantissa	Base	Exponent
3×10^6	3	10	6
110×2^8	110	2	8
6132.784	6132784	10	-3

IEEE 754 floating point number representation

a) Single precision format



b) Double precision format



Q. Represent $(1259.125)_{10}$ in Single and double precision

⇒ Step 1: Convert decimal number to binary.

$$(1259)_{10} = (1001101011)_2$$

$$(0.125)_{10} = (001)_2$$

$$125 \times 2 = 0.250 \times 2 = 0.500 \times 2 = 1.0$$

$$(1259.125)_{10} = (1001101011.001)_2$$

Step 2: Normalize the number

for Single precision $\rightarrow (1.N) 2^{E-127}$

for double precision $\rightarrow (1.N) 2^{E-1023}$

$$1.0011101011.001 = (1.0011101011001 \times 2^{10})$$

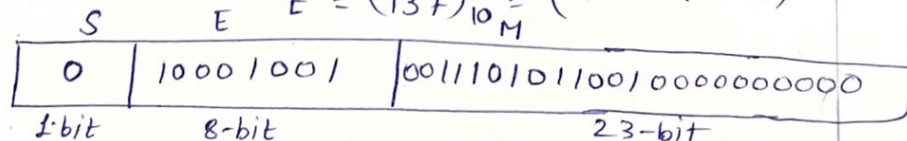
Step 3: Single precision

$$[(1.N) 2^{E-127}]$$

$$(1.0011101011001) \times 2^{10}$$

$$E = 127 = 10$$

$$E = (137)_{10} = (10001001)_M$$



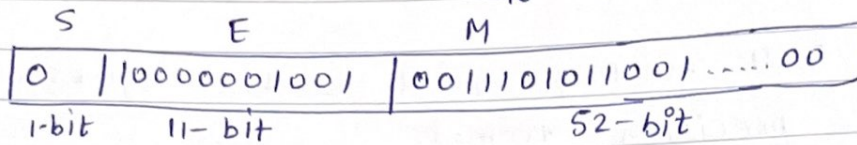
Step 4: Double precision

$$(1.N) 2^{E-1023}$$

$$1.0011101011001 \times 2^{10}$$

$$E-1023 = 10$$

$$E = (1033)_{10} = (100000001001)_2$$



Logic Gates

* Used to represent Boolean function.

- 1) AND
- 2) OR
- 3) NOT
- 4) NAND
- 5) NOR
- 6) X-OR
- 7) X-NOR.